

ch13 Wave Motion

Q6

*6. (a) A laser beam is directed perpendicularly towards a double slit of separation $a = 0.3 \text{ mm}$. The pattern of bright spots projected on a screen 1.8 m away from the slits is shown in Figure 6.1.

Figure 6.1



(i) Find the wavelength of the laser beam.

(3 marks)

(ii) Explain why the slit width has to be very narrow in order for the above pattern to be observed.

(2 marks)

Answers written in the margins will not be marked.

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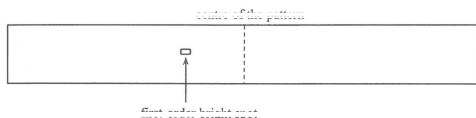
(b) The double slit is now replaced by a diffraction grating of 500 lines per mm .

(i) Find the separation between the central bright spot and first-order bright spot of the pattern on the screen for the same experimental settings.

(3 marks)

(ii) Sketch the pattern, up to the second-order, that you would expect to see on the screen when using this diffraction grating. A first-order bright spot has already been drawn for you.

(2 marks)

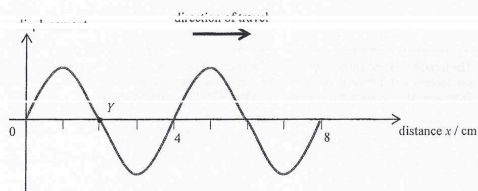


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6. (a) A dipper vibrating with a frequency of 5 Hz is put in a water tank. Figure 6.1 shows the displacement-distance graph of the water wave at time $t = 0$. Y is a particle in the water tank.

Figure 6.1

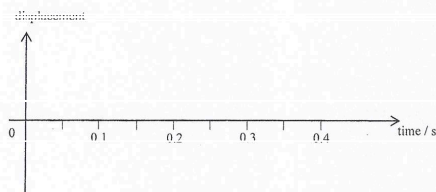


- (i) Determine the wave speed of the water wave. (2 marks)

- (ii) State the direction of motion of particle Y at $t = 0$. (1 mark)

- (iii) Sketch the displacement-time graph of particle Y between $t = 0$ and $t = 0.4$ s in Figure 6.2. (2 marks)

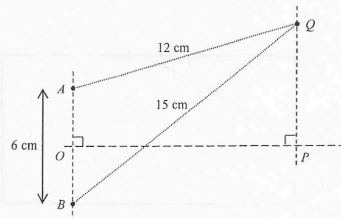
Figure 6.2



Answers written in the margins will not be marked.

- (b) In Figure 6.3, A and B are two dippers vibrating in phase in a water tank. The distance between A and B is 6 cm. OP is the perpendicular bisector of AB . Q is a second minimum from P , where $AQ = 12$ cm and $BQ = 15$ cm.

Figure 6.3

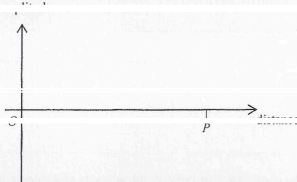


- (i) Explain why a minimum occurs at Q . (2 marks)

- (ii) Determine the wavelength of the water wave. (2 marks)

- (iii) Sketch in Figure 6.4 how the AMPLITUDE of the water wave varies along the line OP . (1 mark)

Figure 6.4



Answers written in the margins will not be marked.

5.

Figure 5.1

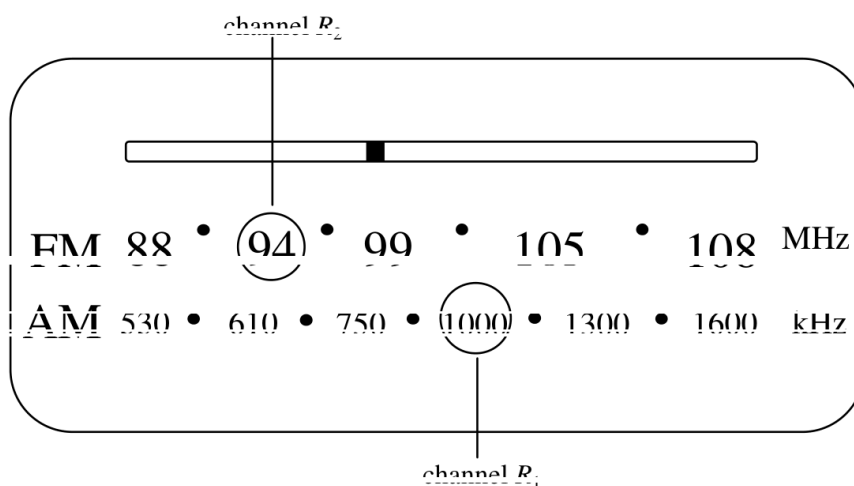


Figure 5.1 shows the display panel of a radio and the broadcasting frequencies of two radio channels R_1 and R_2 . Given : speed of electromagnetic waves = $3.00 \times 10^8 \text{ m s}^{-1}$

- (a) Find the wavelength of the radio waves used by channel R_1 . (1 mark)

- (b) Anita's house is surrounded by hills and at her house, the reception of one of the two radio channels is better. For which radio channel is the reception better? Explain your answer. (2 marks)

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